

Week 5

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1. Complex Numbers

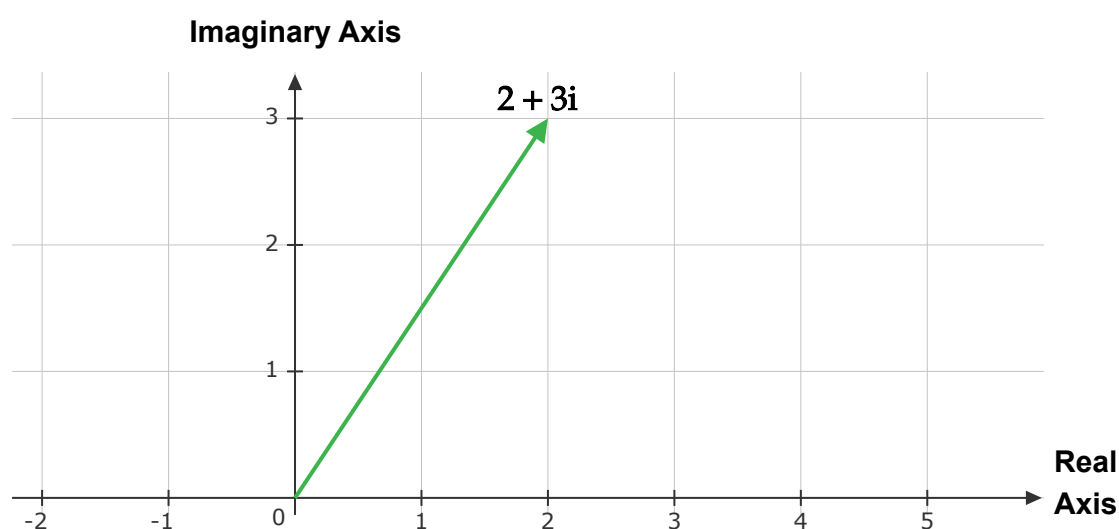
A complex number is of the form $a + ib$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$. The set of all the complex number is given by:

$$\mathbb{C} = \{a + ib | a, b \in \mathbb{R}\}$$

In $a + ib$, a is referred to as the "**real part**" and b is referred to as the "**imaginary part**".

We can plot (a, b) in a **complex plane**. A complex plan has **real axis** and an **imaginary axis**. The horizontal axis is the real axis and the vertical axis is the imaginary axis.

Let us plot $2 + 3i$. The point $(2, 3)$ corresponds to $2 + 3i$.



Let $z = a + ib$. If $b = 0$, then z is a real number. All real numbers are also complex numpex numbers.

The set $\mathbb{R}$ is a proper subset of $\mathbb{C}$.

Addition

$(a + ib) + (c + id)$ is given by:

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

Multiplication

$(a + ib) \times (c + id)$ is given by:

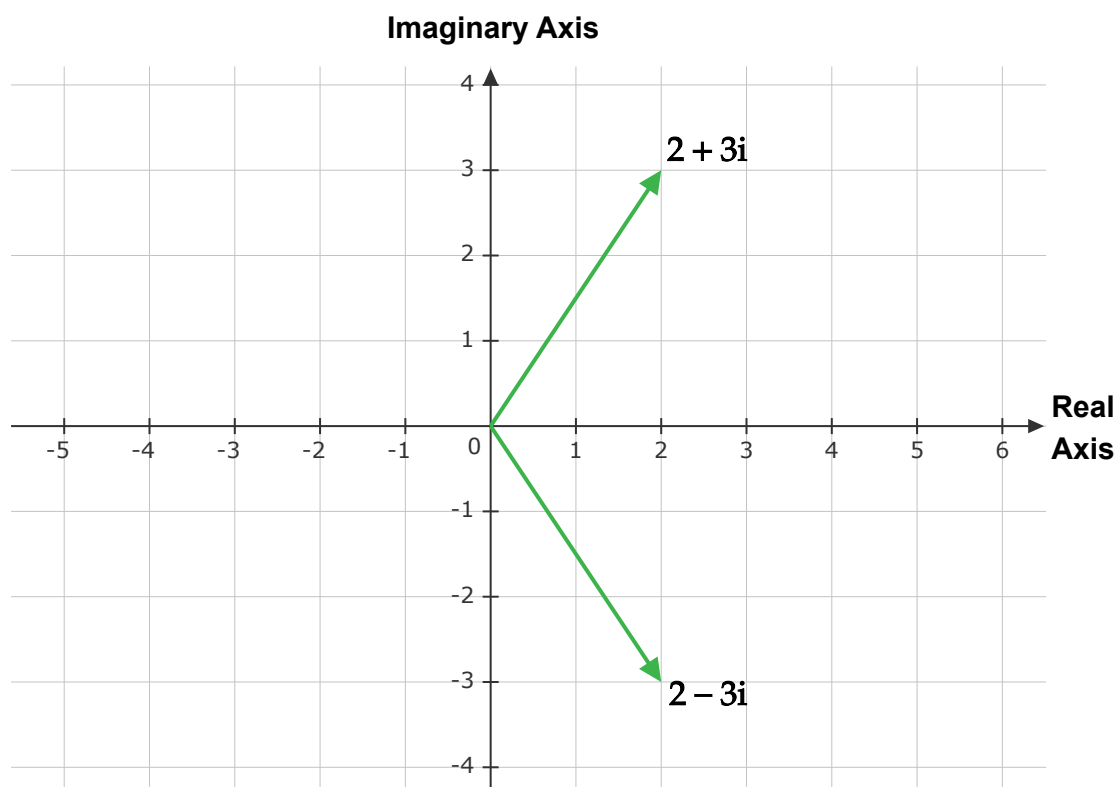
$$(a + ib) \times (c + id) = (ac - bd) + i(ad + bc)$$

Conjugate

Let $z = a + ib$. Then, conjugate of z , denoted by $\bar{z}$, is given by

$$\overline{a + ib} = a - ib$$

Geometrically, the coordinates of the conjugate is a reflection along the real axis.



Let $z = a + ib$, and $\bar{z} = a - ib$. If $z = \bar{z}$, then z is a real number.

$$z = \bar{z}$$

$$a + ib = a - ib$$

$$2ib = 0$$

$$b = 0$$

Since $b = 0$, $a + ib$ is a real number.

Absolute Value

Let $z = a + ib$. The absolute value of z , denoted by $|z|$, is given by

$$\begin{aligned} |a + ib| &= \sqrt{(a + ib)(\overline{a + ib})} \\ &= \sqrt{(a + ib)(a - ib)} \\ &= \sqrt{a^2 + b^2} \end{aligned}$$

Notice that $\sqrt{a^2 + b^2}$ represents the length of the vector (a, b) in the complex plane. The square of the absolute value is given by

$$|z|^2 = z \times \bar{z}$$

The absolute value is always greater than or equal to 0. This is because, it represents the length.

Division

Let $z_1 = a + ib$ and $z_2 = c + id$.

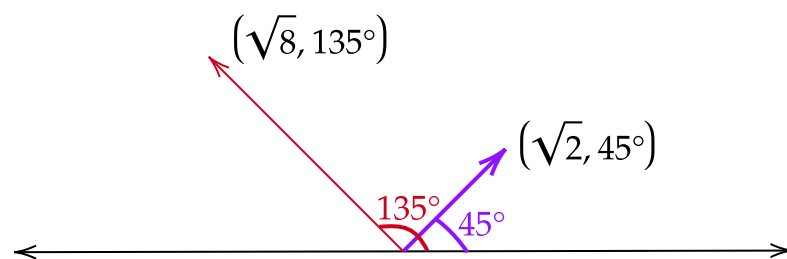
$$\begin{aligned} \frac{z_1}{z_2} &= \frac{z_1 \times \overline{z_2}}{z_2 \times \overline{z_2}} \\ &= \frac{(a + ib) \times (c - id)}{(c + id) \times (c - id)} \\ &= \frac{(ac - bd) + i(ad + bc)}{c^2 + d^2} \\ &= \left(\frac{ac - bd}{c^2 + d^2} \right) + i \left(\frac{ad + bc}{c^2 + d^2} \right) \end{aligned}$$

Polar Coordinates

In the cartesian coordinate system, a point is represented by its x and y coordinates : (x, y) . In the polar coordinate system, a point is represented by its length and its counter clockwise angle with the horizontal axis.

Let us focus on two points.

- $(1, 1)$
 - Length : $\sqrt{2}$
 - Counter clockwise angle with the horizontal axis : 45°
 - Polar Coordinate : $(\sqrt{2}, 45^\circ)$
- $(-2, 2)$
 - Length : $\sqrt{8}$
 - Counter clockwise angle with the horizontal axis : 135°
 - Polar Coordinate : $(\sqrt{8}, 135^\circ)$



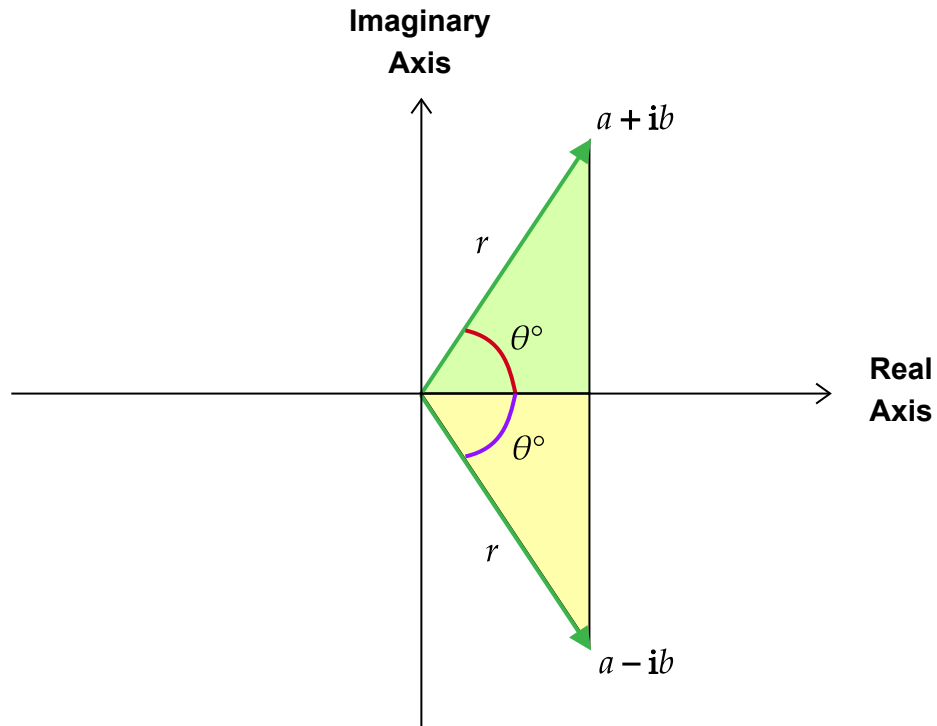
The second coordinate can also be the negative of the clockwise angle with the horizontal axis. For $(1, 1)$, the clockwise angle is 315° . The point $(1, 1)$ can also be expressed as $(\sqrt{2}, -315^\circ)$.

Let us focus on $z = a + ib$ and $\bar{z} = a - ib$, where $a, b \in \mathbb{R}$. Let the length of both the points be r , where

$$r = |z| = |\bar{z}| = \sqrt{a^2 + b^2}$$

Recall, for a right angled triangle, with angle θ° ,

$$\cos(\theta^\circ) = \frac{\text{Length of Adjacent Side}}{\text{Length of the Hypotenuse}}, \quad \sin(\theta^\circ) = \frac{\text{Length of Opposite Side}}{\text{Length of the Hypotenuse}}$$



The following is true for both $a + ib$ and $a - ib$

$$\begin{aligned}\cos(\theta^\circ) &= \frac{a}{r} \\ r \cdot \cos(\theta^\circ) &= a\end{aligned}$$

$$\begin{aligned}\sin(\theta^\circ) &= \frac{b}{r} \\ r \cdot \sin(\theta^\circ) &= b\end{aligned}$$

Recall,

- $\cos(x) = \cos(-x)$
- $-\sin(x) = \sin(-x)$

Euler's Formula :

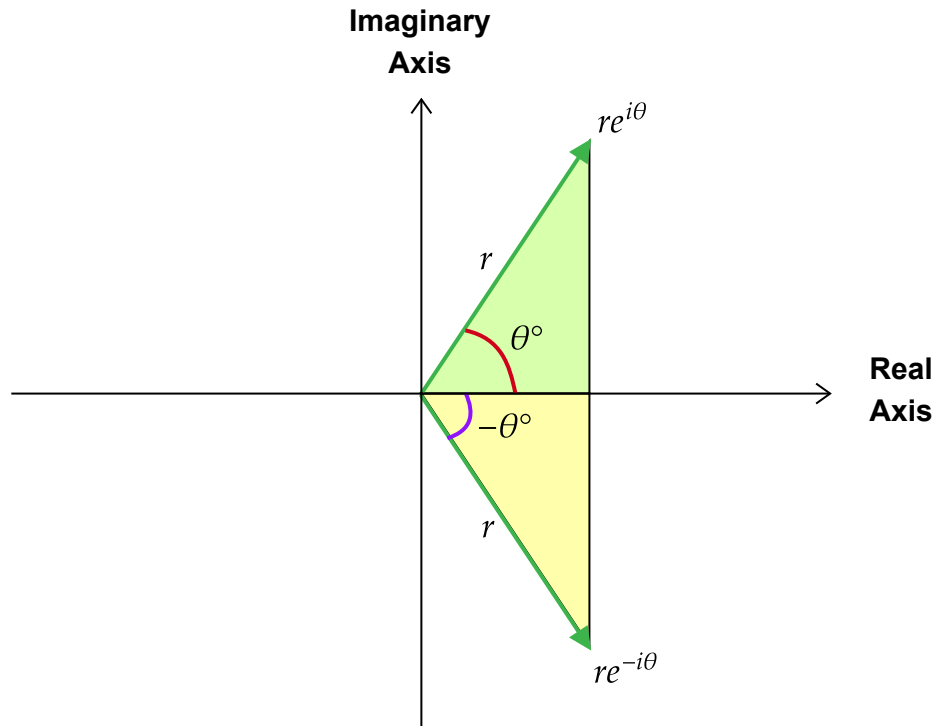
- $e^{ix} = \cos(x) + i \cdot \sin(x)$
- $e^{-ix} = e^{i(-x)} = \cos(-x) + i \cdot \sin(-x)$

Let us substitute these in z and $\bar{z}$:

$$\begin{aligned}z &= a + ib \\ &= r \cdot \cos(\theta^\circ) + i \cdot r \cdot \sin(\theta^\circ) \\ &= r(\cos(\theta^\circ) + i \cdot \sin(\theta^\circ)) \\ &= re^{i\theta}\end{aligned}$$

$$\begin{aligned}\bar{z} &= a - ib \\ &= r \cdot \cos(\theta^\circ) - i \cdot r \cdot \sin(\theta^\circ) \\ &= r(\cos(\theta^\circ) - i \cdot \sin(\theta^\circ)) \\ &= r(\cos(-\theta^\circ) + i \cdot \sin(-\theta^\circ)) \\ &= re^{-i\theta}\end{aligned}$$

Representing z and $\bar{z}$ in the polar coordinate system



2. Complex Matrices

A $n \times 1$ vector u is a complex vector if $u \in \mathbb{C}^n$, where

$$\mathbb{C}^n = \underbrace{\mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}}_{n\text{-times}}$$

Here, $\times$ denotes the cartesian product.

Addition

$$\begin{bmatrix} 1 + 2i \\ 3 + 4i \end{bmatrix} + \begin{bmatrix} 5 + 6i \\ 7 + 8i \end{bmatrix} = \begin{bmatrix} 6 + 8i \\ 10 + 12i \end{bmatrix}$$

Corresponding elements are added.

Scalar Multiplication

$$-2 \times \begin{bmatrix} 1 + 2i \\ 3 + 4i \end{bmatrix} = \begin{bmatrix} -2 - 4i \\ -6 - 8i \end{bmatrix}$$

Every entry is multiplied by the scalar. The scalar can also be a complex number. Example, $(2 + i)$.

$$(2 + i) \times \begin{bmatrix} 1 + 2i \\ 3 + 4i \end{bmatrix} = \begin{bmatrix} 5i \\ 2 + 11i \end{bmatrix}$$

Conjugate

$$\begin{bmatrix} a + ib \\ c + id \end{bmatrix} = \begin{bmatrix} a - ib \\ c - id \end{bmatrix}$$

Conjugate is applied element wise

Transpose

$$A = \begin{bmatrix} a + ib & c + id \\ e + if & g + ih \end{bmatrix}, A^T = \begin{bmatrix} a + ib & e + if \\ c + id & g + ih \end{bmatrix}$$

Conjugate Transpose

Let $A = \begin{bmatrix} a + ib & c + id \\ e + if & g + ih \end{bmatrix}$. The conjugate transpose of A , denoted by $\overline{A}^T$ or A^* is given as follows

$$A^* = \begin{bmatrix} a - ib & e - if \\ c - id & g - ih \end{bmatrix}$$

Properties

Let A, B be $m \times n$ complex matrices, let u, v and w be complex vectors and let $\alpha, \beta \in \mathbb{R}$. Then,

- $(AB)^* = B^* A^*$. This is similar to what we have seen for real matrices P and Q where, $(PQ)^T = Q^T P^T$
- $v^*(\alpha u + \beta w) = \alpha(v^* u) + \beta(v^* w)$
- $((\beta v)^*)(u + w) = \overline{\beta}(v^* u + v^* w)$

Standard Inner Product

Let u and v be two complex vectors. The standard inner product of u and v is given as follows.

$$\langle u, v \rangle = u^* v$$

Note : $u^* v$ need not be equal to $\langle v, u \rangle$. Let $u = \begin{bmatrix} 1 + 2i \\ 3 + 4i \end{bmatrix}$ and let $v = \begin{bmatrix} 5 + 6i \\ 7 + 8i \end{bmatrix}$.

$$\begin{aligned} u^* v &= [1 - 2i \quad 3 - 4i] \times \begin{bmatrix} 5 + 6i \\ 7 + 8i \end{bmatrix} \\ &= (1 - 2i)(5 + 6i) + (3 - 4i)(7 + 8i) \\ &= 5 + 6i - 10i + 12 + 21 + 24i - 28i + 32 \\ &= 70 - 8i \end{aligned}$$

$$v^* u = [5 - 6i \quad 7 - 8i] \times \begin{bmatrix} 1 + 2i \\ 3 + 4i \end{bmatrix}$$

$$\begin{aligned}
&= (5 - 6i)(1 + 2i) + (7 - 8i)(3 + 4i) \\
&= 5 + 10i - 6i + 12 + 21 + 28i - 24i + 32 \\
&= 70 + 8i
\end{aligned}$$

However, $(u^*v) = \overline{(v^*u)}$

$$\begin{aligned}
\overline{(v^*u)} &= \overline{70 + 8i} \\
&= 70 - 8i \\
&= u^*v
\end{aligned}$$

If $u \neq 0$, then $u^*u > 0$. Let $u = \begin{bmatrix} a + ib \\ c + id \end{bmatrix}$

$$\begin{aligned}
u^*u &= [a - ib \quad c - id] \times \begin{bmatrix} a + ib \\ c + id \end{bmatrix} \\
&= (a - ib)(a + ib) + (c - id)(c + id) \\
&= a^2 + b^2 + c^2 + d^2 \\
&> 0
\end{aligned}$$

Since $u \neq 0$, at least one of a, b, c and d is not zero. For any u , u^*u is a real number.

- The **length** of a complex vector u is given by $||u|| = \sqrt{u^*u}$
- If $u^*v = 0$, then u and v are orthogonal to each other.

3. Hermitian Matrix

A complex square matrix A is a hermitian matrix if $A^* = A$.

Example

$$A = \begin{bmatrix} 1 & 3 + 4i \\ 3 - 4i & -5 \end{bmatrix}, \quad A^* = \begin{bmatrix} 1 & 3 + 4i \\ 3 - 4i & -5 \end{bmatrix}$$

- The diagonal elements of a hermitian matrix are always real numbers.

Property 1 : The eigenvalues of a hermitian matrix are real.

Proof : Let $A = A^*$ and let $Av = \lambda v$, where $v \neq 0$.

$$\begin{aligned}
\lambda v^* v &= v^* (\lambda v) \\
\lambda v^* v &= v^* (Av) \\
\lambda v^* v &= v^* A^* v \\
\lambda v^* v &= (Av)^* v \\
\lambda v^* v &= (\lambda v)^* v \\
\lambda v^* v &= \bar{\lambda} (v^* v) \\
\lambda v^* v - \bar{\lambda} v^* v &= 0 \\
(\lambda - \bar{\lambda})(v^* v) &= 0 \\
\lambda &= \bar{\lambda}
\end{aligned}$$

Since v is an eigenvector, $v \neq 0$. Therefore, $v^* v \neq 0$. Since $\lambda = \bar{\lambda}$, we can say that λ is a real number.

Property 2 : The eigenvectors corresponding to distinct eigenvalues of a hermitian matrix are orthogonal.

Proof : Let $A^* = A$, $Av = \lambda_1 v$ and $Au = \lambda_2 u$, where $v \neq 0$ and $u \neq 0$ and let $\lambda_1 \neq \lambda_2$.

$$\begin{aligned}
Av &= \lambda_1 v \\
u^* Av &= \lambda_1 (u^* v) \\
u^* A^* v &= \lambda_1 (u^* v) \\
(Au)^* v &= \lambda_1 (u^* v) \\
(\lambda_2 u)^* v &= \lambda_1 (u^* v) \\
\bar{\lambda}_2 (u^* v) &= \lambda_1 (u^* v) \\
\lambda_2 (u^* v) &= \lambda_1 (u^* v) \\
(\lambda_2 - \lambda_1)(u^* v) &= 0 \\
u^* v &= 0
\end{aligned}$$

The eigenvalues of a hermitian matrix are real numbers. Therefore, $\bar{\lambda}_2 = \lambda_2$. Since $\lambda_1 \neq \lambda_2$, $\lambda_2 - \lambda_1 \neq 0$.

Real Symmetric Matrices

A real symmetric matrix is a hermitian matrix. Therefore,

- All the eigenvalues of a real symmetric matrix are real.
 - The eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal
-

4. Unitary Matrix

A complex square matrix U is unitary if $U^*U = I = UU^*$.

- The rows of a unitary matrix form an orthonormal set.
- The columns of a unitary matrix form an orthonormal set.

In fact, for a square matrix, if one of the above two bullet points hold true, then that matrix is a unitary matrix.

Example

$$U = \begin{bmatrix} 1/\sqrt{2} & i/\sqrt{2} \\ i/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Let us denote the first and second columns as c_1 and c_2 . The length of c_1 and c_2 are given as follows:

$$\begin{aligned} \sqrt{c_1^* c_1} &= \sqrt{\begin{bmatrix} 1/\sqrt{2} & -i/\sqrt{2} \end{bmatrix} \times \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix}} & \sqrt{c_2^* c_2} &= \sqrt{\begin{bmatrix} -i/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \times \begin{bmatrix} i/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}} \\ &= \sqrt{(1/\sqrt{2})^2 + (1/\sqrt{2})^2} & &= \sqrt{(1/\sqrt{2})^2 + (1/\sqrt{2})^2} \\ &= \sqrt{(1/2) + (1/2)} & &= \sqrt{(1/2) + (1/2)} \\ &= \sqrt{1} & &= \sqrt{1} \\ &= 1 & &= 1 \end{aligned}$$

The inner product of c_1 and c_2 is given as

$$\begin{aligned} c_1^* c_2 &= \begin{bmatrix} 1/\sqrt{2} & -i/\sqrt{2} \end{bmatrix} \times \begin{bmatrix} i/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \\ &= (i/2) + (-i/2) \\ &= 0 \end{aligned}$$

Therefore, U is a unitary matrix.

Property 1 : If $U_{n \times n}$ is a unitary matrix, then $\|Uv\|^2 = \|v\|^2$ for all $v \in \mathbb{C}^n$.

Proof :

$$\begin{aligned} \|Uv\|^2 &= (Uv)^*(Uv) \\ &= v^* U^* U v \end{aligned}$$

$$\begin{aligned}
&= v^* I v \\
&= v^* v \\
&= ||v||^2
\end{aligned}$$

When a vector is multiplied with a unitary matrix, the length is unchanged.

Property 2 : The absolute value of all the eigenvalues of a unitary matrix is 1

Proof : Let U be a unitary matrix and let $Uv = \lambda v$, where $v \neq 0$.

$$\begin{aligned}
||Uv||^2 &= ||v||^2 \\
(Uv)^*(Uv) &= v^* v \\
(\lambda v)^*(\lambda v) &= v^* v \\
(\bar{\lambda} \lambda)(v^* v) &= v^* v \\
(\bar{\lambda} \lambda - 1)(v^* v) &= 0 \\
\bar{\lambda} \lambda - 1 &= 0 \\
\bar{\lambda} \lambda &= 1 \\
|\lambda|^2 &= 1 \\
|\lambda| &= 1
\end{aligned}$$

Since $v \neq 0$, $v^* v \neq 0$. The absolute value of a complex number is always greater than or equal to 0, therefore, $|\lambda| = 1$.

Note : The absolute value being equal to 1 doesn't imply that all the eigenvalues of a unitary matrix are either 1 or -1 . For example $\lambda = \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$ can be an eigenvalue of a unitary matrix.

$$|\lambda| = \sqrt{\lambda \times \bar{\lambda}} = \sqrt{\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right) \times \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)} = \sqrt{\left(\frac{1}{2}\right) + \left(\frac{i}{2}\right) - \left(\frac{i}{2}\right) + \left(\frac{1}{2}\right)} = \sqrt{1} = 1$$

Property 3 : The eigenvectors corresponding to distinct eigenvalues of a unitary matrix are orthogonal.

Proof : Let $U_{n \times n}$ be a unitary matrix. Let $Uv = \lambda_1 v$ and $Uw = \lambda_2 w$, where $v \neq 0$, $w \neq 0$ and $\lambda_1 \neq \lambda_2$.

$$\begin{aligned}
v^* w &= v^* I w \\
v^* w &= v^* U^* U w \\
v^* w &= (Uv)^*(Uw)
\end{aligned}$$

$$\begin{aligned}
v^*w &= (\lambda_1 v)^*(\lambda_2 w) \\
v^*w &= (\overline{\lambda_1} \lambda_2) (v^*w) \\
(\overline{\lambda_1} \lambda_2 - 1) (v^*w) &= 0
\end{aligned}$$

If we can show that $\overline{\lambda_1} \lambda_2 - 1 \neq 0$, then we can say that $v^*w = 0$. Let us assume that $\overline{\lambda_1} \lambda_2 - 1 = 0$.

$$\begin{aligned}
\overline{\lambda_1} \lambda_2 - 1 &= 0 \\
\overline{\lambda_1} \lambda_2 &= 1 \\
\lambda_1 \overline{\lambda_1} \lambda_2 &= \lambda_1 \\
|\lambda_1|^2 \lambda_2 &= \lambda_1 \\
(1) \lambda_2 &= \lambda_1 \\
\lambda_2 &= \lambda_1
\end{aligned}$$

The absolute value of the eigenvalues of a unitary matrix is 1. Therefore, $|\lambda_1|^2 = 1$. However, we see that if $\overline{\lambda_1} \lambda_2 - 1 = 0$, then $\lambda_1 = \lambda_2$. This contradicts the fact that $\lambda_1 \neq \lambda_2$. Our assumption is wrong. $\overline{\lambda_1} \lambda_2 - 1 \neq 0$. Therefore, $v^*w = 0$.

Property 4 : Let $U = U_1 U_2$, where U_1 and U_2 are unitary matrices. U is also a unitary matrix.

Proof :

$$\begin{aligned}
U^*U &= (U_1 U_2)^*(U_1 U_2) \\
&= U_2^* U_1^* U_1 U_2 \\
&= U_2^* I U_2 \\
&= I
\end{aligned}$$

Product of two unitary matrices is also a unitary matrix.

Orthogonal Matrix

An orthogonal matrix is a unitary matrix. Let $Q_{n \times n}$ be an orthogonal matrix.

- $Q^T Q = I = Q Q^T$
- The rows of an orthogonal matrix form an orthonormal set.
- The columns of a orthogonal matrix form an orthonormal set.
- $\|Qv\|^2 = \|v\|^2$, for all $v \in \mathbb{R}^n$
- The absolute value of all the eigenvalues of Q is 1
- The eigenvectors corresponding to distinct eigenvalues of Q are orthogonal.
- Product of two orthogonal matrices is also an orthogonal matrix.

Once again, the eigenvalues of Q can be complex. Example,

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The eigenvalues of the above matrix are i and $-i$.

5. Block Matrix Multiplication

Consider matrix X

$$X = \begin{bmatrix} a_1 & a_2 & b_1 & b_2 & b_3 \\ a_3 & a_4 & b_4 & b_5 & b_6 \\ c_1 & c_2 & d_1 & d_2 & d_3 \\ c_3 & c_4 & d_4 & d_5 & d_6 \\ c_5 & c_6 & d_7 & d_8 & d_9 \end{bmatrix}$$

The first row to fifth row of X are denoted by R_1, R_2, R_3, R_4 and R_5 respectively

Consider matrix Y

$$Y = \begin{bmatrix} p_1 & p_2 & q_1 & q_2 & q_3 \\ p_3 & p_4 & q_4 & q_5 & q_6 \\ r_1 & r_2 & s_1 & s_2 & s_3 \\ r_3 & r_4 & s_4 & s_5 & s_6 \\ r_5 & r_6 & s_7 & s_8 & s_9 \end{bmatrix}$$

The first column to fifth column of Y are denoted by C_1, C_2, C_3, C_4 and C_5 respectively

Consider the following matrices

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \end{bmatrix}, C = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \\ c_5 & c_6 \end{bmatrix}, D = \begin{bmatrix} d_1 & d_2 & d_3 \\ d_4 & d_5 & d_6 \\ d_7 & d_8 & d_9 \end{bmatrix}$$

$$P = \begin{bmatrix} p_1 & p_2 \\ p_3 & p_4 \end{bmatrix}, Q = \begin{bmatrix} q_1 & q_2 & q_3 \\ q_4 & q_5 & q_6 \end{bmatrix}, R = \begin{bmatrix} r_1 & r_2 \\ r_3 & r_4 \\ r_5 & r_6 \end{bmatrix}, S = \begin{bmatrix} s_1 & s_2 & s_3 \\ s_4 & s_5 & s_6 \\ s_7 & s_8 & s_9 \end{bmatrix}$$

Matrix X and Y in block form:

$$X = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, Y = \begin{bmatrix} P & Q \\ R & S \end{bmatrix}$$

To show:

$$XY = \begin{bmatrix} AP + BR & AQ + BS \\ CP + DR & CQ + DS \end{bmatrix}$$

$$\begin{bmatrix} a_1 & a_2 & b_1 & b_2 & b_3 \\ a_3 & a_4 & b_4 & b_5 & b_6 \\ c_1 & c_2 & d_1 & d_2 & d_3 \\ c_3 & c_4 & d_4 & d_5 & d_6 \\ c_5 & c_6 & d_7 & d_8 & d_9 \end{bmatrix} \begin{bmatrix} p_1 & p_2 & q_1 & q_2 & q_3 \\ p_3 & p_4 & q_4 & q_5 & q_6 \\ r_1 & r_2 & s_1 & s_2 & s_3 \\ r_3 & r_4 & s_4 & s_5 & s_6 \\ r_5 & r_6 & s_7 & s_8 & s_9 \end{bmatrix} = \begin{bmatrix} (AP + BR)_{2 \times 2} & (AQ + BS)_{2 \times 3} \\ (CP + DR)_{3 \times 2} & (CQ + DS)_{3 \times 3} \end{bmatrix}_{5 \times 5}$$

Proof :

XY is given as

$$XY = \begin{bmatrix} R_1 \cdot C_1 & R_1 \cdot C_2 & R_1 \cdot C_3 & R_1 \cdot C_4 & R_1 \cdot C_5 \\ R_2 \cdot C_1 & R_2 \cdot C_2 & R_2 \cdot C_3 & R_2 \cdot C_4 & R_2 \cdot C_5 \\ R_3 \cdot C_1 & R_3 \cdot C_2 & R_3 \cdot C_3 & R_3 \cdot C_4 & R_3 \cdot C_5 \\ R_4 \cdot C_1 & R_4 \cdot C_2 & R_4 \cdot C_3 & R_4 \cdot C_4 & R_4 \cdot C_5 \\ R_5 \cdot C_1 & R_5 \cdot C_2 & R_5 \cdot C_3 & R_5 \cdot C_4 & R_5 \cdot C_5 \end{bmatrix}$$

Let us find AP, BR

$$AP = \begin{bmatrix} a_1 p_1 + a_2 p_3 & a_1 p_2 + a_2 p_4 \\ a_3 p_1 + a_4 p_3 & a_3 p_2 + a_4 p_4 \end{bmatrix} \quad BR = \begin{bmatrix} b_1 r_1 + b_2 r_3 + b_3 r_5 & b_1 r_2 + b_2 r_4 + b_3 r_6 \\ b_4 r_1 + b_5 r_3 + b_6 r_5 & b_4 r_2 + b_5 r_4 + b_6 r_6 \end{bmatrix}$$

$$AP + BR = \begin{bmatrix} a_1 p_1 + a_2 p_3 + b_1 r_1 + b_2 r_3 + b_3 r_5 & a_1 p_2 + a_2 p_4 + b_1 r_2 + b_2 r_4 + b_3 r_6 \\ a_3 p_1 + a_4 p_3 + b_4 r_1 + b_5 r_3 + b_6 r_5 & a_3 p_2 + a_4 p_4 + b_4 r_2 + b_5 r_4 + b_6 r_6 \end{bmatrix}$$

Notice that $AP + BR = \begin{bmatrix} R_1 \cdot C_1 & R_1 \cdot C_2 \\ R_2 \cdot C_1 & R_2 \cdot C_2 \end{bmatrix}$

Similarly,

$$AQ + BS = \begin{bmatrix} R_1 \cdot C_3 & R_1 \cdot C_4 & R_1 \cdot C_5 \\ R_2 \cdot C_3 & R_2 \cdot C_4 & R_2 \cdot C_5 \end{bmatrix}$$

$$CP + DR = \begin{bmatrix} R_3 \cdot C_1 & R_3 \cdot C_2 \\ R_4 \cdot C_1 & R_4 \cdot C_2 \\ R_5 \cdot C_1 & R_5 \cdot C_2 \end{bmatrix}$$

$$CQ + DS = \begin{bmatrix} R_3 \cdot C_3 & R_3 \cdot C_4 & R_3 \cdot C_5 \\ R_4 \cdot C_3 & R_4 \cdot C_4 & R_4 \cdot C_5 \\ R_5 \cdot C_3 & R_5 \cdot C_4 & R_5 \cdot C_5 \end{bmatrix}$$

Therefore,

$$XY = \begin{bmatrix} AP + BR & AQ + BS \\ CP + DR & CQ + DS \end{bmatrix}$$

6. Schur's Theorem

For every square matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary matrix U and an upper triangular matrix T such that,

$$A = UTU^*$$

Proof:

Suppose, if $A = [a + bi]_{1 \times 1}$, where $a, b \in \mathbb{R}$, then $U = [1]$, $U^* = [1]$ and $T = [a + ib]$.

$$A = UTU^*$$

T is an upper triangular matrix. Therefore, Schur's theorem holds for 1×1 matrices.

Assume : Any $(n - 1) \times (n - 1)$ matrix can be decomposed as UTU^* , where $n > 1$.

Let A be an $n \times n$ matrix. Let $Av_1 = \lambda v_1$, where $\|v_1\| = 1$. Let Q_1 be an $n \times n$ unitary matrix.

$$Q_1 = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$$

Here, v_1 is an eigenvector of A . $v_2, \dots, v_n$ need not be the eigenvectors of A . We can obtain $v_2, \dots, v_n$ using gram schmidt process. Let us expand $Q_1^* A Q_1$

$$\begin{aligned} Q_1^* A Q_1 &= \begin{bmatrix} - & v_1^* & - \\ - & v_2^* & - \\ & \vdots & \\ - & v_n^* & - \end{bmatrix} A \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \\ &= \begin{bmatrix} - & v_1^* & - \\ - & v_2^* & - \\ & \vdots & \\ - & v_n^* & - \end{bmatrix} \begin{bmatrix} \lambda v_1 & Av_2 & \cdots & Av_n \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} \lambda v_1^* v_1 & v_1^* A v_2 & \cdots & v_1^* A v_n \\ \lambda v_2^* v_1 & v_2^* A v_2 & \cdots & v_2^* A v_n \\ \vdots & \vdots & \ddots & \vdots \\ \lambda v_n^* v_1 & v_n^* A v_2 & \cdots & v_n^* A v_n \end{bmatrix} \\
&= \begin{bmatrix} \lambda & v_1^* A v_2 & \cdots & v_1^* A v_n \\ 0 & v_2^* A v_2 & \cdots & v_2^* A v_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & v_n^* A v_2 & \cdots & v_n^* A v_n \end{bmatrix}
\end{aligned}$$

- Since $\|v_1\| = 1$, $v_1^* v_1 = 1$.
- $\lambda v_2^* v_1 = 0$ as v_2 and v_1 are orthogonal. Similarly, $\lambda v_n^* v_1 = 0$

Consider the following,

$$w = [v_1^* A v_2 \quad \cdots \quad v_1^* A v_n], \quad 0 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \quad B = \begin{bmatrix} v_2^* A v_2 & \cdots & v_2^* A v_n \\ \vdots & \ddots & \vdots \\ v_n^* A v_2 & \cdots & v_n^* A v_n \end{bmatrix}$$

- w is a $1 \times (n-1)$ row vector.
- 0 is an $(n-1) \times 1$ column vector with all the entries equal to 0.
- B is an $(n-1) \times (n-1)$ matrix.

The block matrix form of $Q_1^* A Q_1$ is given as follows

$$\begin{aligned}
Q_1^* A Q_1 &= \begin{bmatrix} \lambda & v_1^* A v_2 & \cdots & v_1^* A v_n \\ 0 & v_2^* A v_2 & \cdots & v_2^* A v_n \\ \vdots & \vdots & \ddots & \vdots \\ 0 & v_n^* A v_2 & \cdots & v_n^* A v_n \end{bmatrix} \\
&= \begin{bmatrix} \lambda & w \\ 0 & B \end{bmatrix}
\end{aligned}$$

By our assumption, we can decompose B as follows

$$B = Q_2 T_2 Q_2^*$$

Where, Q_2 is a unitary matrix and T_2 is an upper triangular matrix.

Let $z = w Q_2$. Then $w = z Q_2^*$. Then,

$$Q_1^* A Q_1 = \begin{bmatrix} \lambda & w \\ 0 & B \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} \lambda & zQ_2^* \\ 0 & Q_2 T_2 Q_2^* \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & Q_2 \end{bmatrix} \times \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & Q_2^* \end{bmatrix}
\end{aligned}$$

Let $\widehat{Q} = \begin{bmatrix} 1 & 0 \\ 0 & Q_2 \end{bmatrix}$. Then, $\widehat{Q}^* = \begin{bmatrix} 1 & 0 \\ 0 & Q_2^* \end{bmatrix}$. $\widehat{Q}$ is also a unitary matrix.

$$\begin{aligned}
Q_1^* A Q_1 &= \begin{bmatrix} 1 & 0 \\ 0 & Q_2 \end{bmatrix} \times \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & Q_2^* \end{bmatrix} \\
&= \widehat{Q} \times \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix} \times \widehat{Q}^*
\end{aligned}$$

Let $U = Q_1 \widehat{Q}$. U is also a unitary matrix. $U^* = \widehat{Q}^* Q_1^*$. Let us expand $U^* A U$

$$U^* A U = \widehat{Q}^* Q_1^* A Q_1 \widehat{Q}$$

Since $Q_1^* A Q_1 = \widehat{Q} \times \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix} \times \widehat{Q}^*$, we get the following.

$$\begin{aligned}
U^* A U &= \widehat{Q}^* Q_1^* A Q_1 \widehat{Q} \\
&= \widehat{Q}^* \widehat{Q} \times \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix} \times \widehat{Q}^* \widehat{Q} \\
&= \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix}
\end{aligned}$$

Let $T = \begin{bmatrix} \lambda & z \\ 0 & T_2 \end{bmatrix}$. T is an upper triangular matrix.

$$\begin{aligned}
U^* A U &= T \\
A &= U T U^*
\end{aligned}$$

7. Spectral Theorem

Let A be a hermitian matrix. Then, A is unitarily diagonalizable.

$$A = UDU^*$$

Here,

- D is a diagonal matrix, whose diagonal entries are the eigenvalues of A
- U is a unitary matrix. The columns of U are the eigenvectors of A whose corresponding eigenvalues are the diagonal entries of D .

Proof : By Schur's theorem, there exists a unitary matrix Q and an upper triangular matrix T such that,

$$A = QTQ^*$$

$$A^* = (QTQ^*)^*$$

$$A^* = QT^*Q^*$$

Since A is a hermitian matrix,

$$A^* = A$$

$$QT^*Q^* = QTQ^*$$

$$T^* = T$$

Since T is an upper triangular matrix, T^* is a lower triangular matrix. An upper triangular matrix is equal to its conjugate transpose if and only if it is a diagonal matrix. Therefore, T is a diagonal matrix.

Real Symmetric Matrix & Orthogonally Diagonalizable

A real symmetric matrix S is also a hermitian matrix. Therefore, S is unitarily diagonalizable.

$$S = QDQ^*$$

Here,

- D is a diagonal matrix, whose diagonal entries are the eigenvalues of S
- Q is a unitary matrix. The columns of Q are the eigenvectors of S whose corresponding eigenvalues are the diagonal entries of D .

Since S is a real symmetric matrix, we can further say that Q is an orthogonal matrix.

Proof :

Let $v \neq 0$ be an eigenvector of $S_{n \times n}$, i.e., $Sv = \lambda v$.

Let $a, b \in \mathbb{R}^n$ such that, $v = a + ib$.

$$Sv = \lambda v$$

$$S(a + ib) = \lambda(a + ib)$$

$$Sa + i(Sb) = \lambda a + i(\lambda b)$$

By equating the real and imaginary part, we get

$$Sa = \lambda a$$

$$Sb = \lambda b$$

a and b are also eigenvectors of S with λ as eigenvalues. This means, for any eigenvalue of S , we can a real eigenvector. Therefore, we can construct an orthogonal matrix Q .
